MTH 201 Chapter 5: Parametric/Polar Curves & Conics

5.3.1 Tangent Lines & Arc Lengths of Polar Curves

Tangent Lines of Polar Curves

Motivation: Consider an arbitrary polar curve C of the form $r = f(\theta)$. Suppose we want to find the equation of a tangent line to some point on the curve.

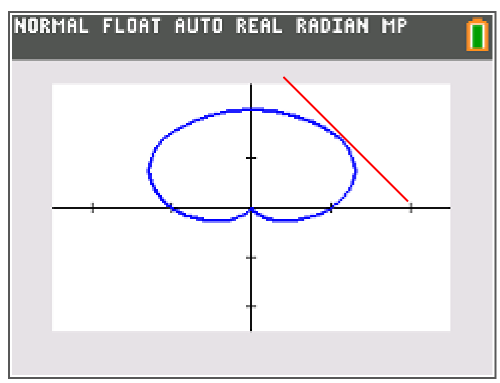
To do this, we will have to calculate $\frac{dy}{dx}$. This will be possible if we view θ as a parameter and note that the curve C is also given by the parametric equations: $x = r\cos(\theta) = f(\theta)\cos(\theta) \quad y = r\sin(\theta) = f(\theta)\sin(\theta)$. Then, we can use the methods of section 7.2 to calculate $\frac{dy}{dx}$.

Theorem: In order to find the slope of a tangent line on a polar curve at a given angle θ we calculate:

$$\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)} = \frac{f'(\theta)\sin(\theta) + f(\theta)\cos(\theta)}{f'(\theta)\cos(\theta) - f(\theta)\sin(\theta)} \quad \textit{provided that } \frac{dx}{d\theta} \neq 0$$

Note: The product rule is being used to find $\frac{dy}{d\theta}$ *and* $\frac{dx}{d\theta}$.

Example 1: For the cardioid $r = 1 + \sin(\theta)$ find the slope of the tangent line at the point $\left(1 + \frac{\sqrt{3}}{2}, \frac{\pi}{3}\right)$.



We have that $x = (1 + \sin(\theta))\cos(\theta)$ and $y = (1 + \sin(\theta))\sin(\theta)$.

Thus, we have that $\frac{dx}{d\theta} = -(1 + \sin(\theta))\sin(\theta) + \cos^2(\theta)$ and

$\frac{dy}{d\theta} = (1 + \sin(\theta))\cos(\theta) + \cos(\theta)\sin(\theta)$. We note that $\left.\frac{dx}{d\theta}\right|_{\theta = \frac{\pi}{3}} \neq 0$.

Thus, the answer is:

$$\left.\frac{dy}{dx}\right|_{\theta = \frac{\pi}{3}} = \left.\frac{(1 + \sin(\theta))\cos(\theta) + \cos(\theta)\sin(\theta)}{-(1 + \sin(\theta))\sin(\theta) + \cos^2(\theta)}\right|_{\theta = \frac{\pi}{3}} = \frac{\left(1 + \frac{\sqrt{3}}{2}\right)\frac{1}{2} + \frac{1}{2}\frac{\sqrt{3}}{2}}{-\left(1 + \frac{\sqrt{3}}{2}\right)\frac{\sqrt{3}}{2} + \frac{1}{4}} = \frac{\frac{\sqrt{3}}{2} + \frac{1}{2}}{-\frac{1}{2} - \frac{\sqrt{3}}{2}} = -1.$$

Example 2: Find the points on the cardioid $r = 1 + \sin(\theta)$, on the interval $[0, 2\pi)$ where there are horizontal & vertical tangent lines and singular points.

Remember: $\begin{aligned} x &= f(\theta)\cos(\theta) = [1+\sin(\theta)]\cos(\theta) \\ y &= f(\theta)\sin(\theta) = [1+\sin(\theta)]\sin(\theta) \end{aligned}$. We must first find when $\dfrac{dx}{d\theta} = 0$ & $\dfrac{dy}{d\theta} = 0$.

$$\begin{aligned} \frac{dx}{d\theta} &= \cos^2(\theta) - [1+\sin(\theta)]\sin(\theta) \\ &= [1-\sin^2(\theta)] - [1+\sin(\theta)]\sin(\theta) \\ &= 1 - \sin(\theta) - 2\sin^2(\theta) \\ &= [1-2\sin(\theta)][1+\sin(\theta)] \end{aligned}$$

So we need $[1-2\sin(\theta)][1+\sin(\theta)] = 0$

$$\begin{array}{lcl} 1-2\sin(\theta)=0 & \text{or} & 1+\sin(\theta)=0 \\ \sin(\theta)=\dfrac{1}{2} & \text{or} & \sin(\theta)=-1 \\ \theta=\dfrac{\pi}{6},\dfrac{5\pi}{6} & \text{or} & \theta=\dfrac{3\pi}{2} \end{array}$$

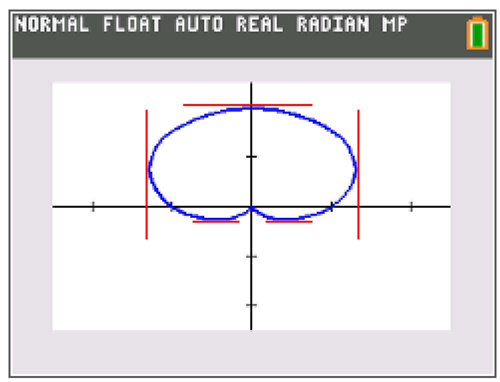
$$\begin{aligned} \frac{dy}{d\theta} &= \cos(\theta)\sin(\theta) + [1+\sin(\theta)]\cos(\theta) \\ &= \cos(\theta)[\sin(\theta)+1+\sin(\theta)] \\ &= \cos(\theta)[2\sin(\theta)+1] \end{aligned}$$

So we need $\cos(\theta)[2\sin(\theta)+1] = 0$

$$\begin{array}{lcl} \cos(\theta)=0 & \text{or} & 2\sin(\theta)+1=0 \\ \cos(\theta)=0 & \text{or} & \sin(\theta)=-\dfrac{1}{2} \\ \theta=\dfrac{\pi}{2},\dfrac{3\pi}{2} & \text{or} & \theta=\dfrac{7\pi}{6},\dfrac{11\pi}{6} \end{array}$$

Horizontal tangent lines: $\left(\dfrac{dy}{d\theta} = 0 \text{ \& } \dfrac{dx}{d\theta} \neq 0\right)$

There are horizontal tangents at: $\left(2, \dfrac{\pi}{2}\right)$, $\left(\dfrac{1}{2}, \dfrac{7\pi}{6}\right)$, & $\left(\dfrac{1}{2}, \dfrac{11\pi}{6}\right)$



Vertical tangent line $\left(\dfrac{dx}{d\theta} = 0 \text{ \& } \dfrac{dy}{d\theta} \neq 0\right)$

There are vertical tangents at: $\left(\dfrac{3}{2}, \dfrac{\pi}{6}\right)$ & $\left(\dfrac{3}{2}, \dfrac{5\pi}{6}\right)$

Singular points $\left(\dfrac{dy}{d\theta} = 0 \text{ \& } \dfrac{dx}{d\theta} = 0\right)$

There is a singular point at: $\left(0, \dfrac{3\pi}{2}\right)$

Note: We could show, through evaluation of limits, that there is a vertical tangent line at this point as well. Since details of this are omitted in the textbook we will also omit the details and just consider this as a "singular point".

Example 3 (you try): Find the points on the curve given by $r = 1-\cos(\theta)$ from $0 \le \theta < 2\pi$ where there are horizontal & vertical tangent lines and singular points.

Remember: $\begin{aligned} x &= f(\theta)\cos(\theta) = [1-\cos(\theta)]\cos(\theta) \\ y &= f(\theta)\sin(\theta) = [1-\cos(\theta)]\sin(\theta) \end{aligned}$. We must first find when $\dfrac{dx}{d\theta} = 0$ & $\dfrac{dy}{d\theta} = 0$.

$$\frac{dx}{d\theta} = \sin(\theta)\cos(\theta) - [1-\cos(\theta)]\sin(\theta)$$
$$= 2\sin(\theta)\cos(\theta) - \sin(\theta)$$
$$= \sin(\theta)[2\cos(\theta)-1]$$

So we need $\sin(\theta)[2\cos(\theta)-1] = 0$

$\sin(\theta) = 0$ or $2\cos(\theta) - 1 = 0$

$\cos(\theta) = \dfrac{1}{2}$

$\theta = 0, \pi$ or $\theta = \dfrac{\pi}{3}, \dfrac{5\pi}{3}$

$$\frac{dy}{d\theta} = \sin^2(\theta) + [1-\cos(\theta)]\cos(\theta)$$
$$= \sin^2(\theta) + \cos(\theta) - \cos^2(\theta)$$
$$= 1 + \cos(\theta) - 2\cos^2(\theta)$$
$$= [-2\cos(\theta)-1][\cos(\theta)-1]$$

So we need $[-2\cos(\theta)-1][\cos(\theta)-1] = 0$

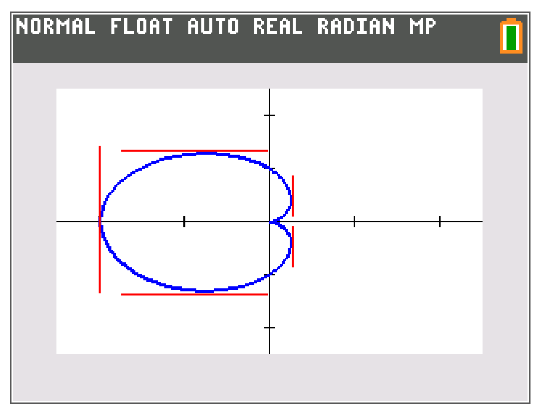
$-2\cos(\theta) - 1 = 0$ or $\cos(\theta) - 1 = 0$

$\cos(\theta) = -\dfrac{1}{2}$ or $\cos(\theta) = 1$

$\theta = \dfrac{2\pi}{3}, \dfrac{4\pi}{3}$ or $\theta = 0$

Horizontal tangent lines: $\left(\dfrac{dy}{d\theta} = 0 \text{ \& } \dfrac{dx}{d\theta} \ne 0\right)$

There are horizontal tangents at: $\left(\dfrac{3}{2}, \dfrac{2\pi}{3}\right)$ & $\left(\dfrac{3}{2}, \dfrac{4\pi}{3}\right)$



Vertical tangent line $\left(\dfrac{dx}{d\theta} = 0 \text{ \& } \dfrac{dy}{d\theta} \ne 0\right)$

There are vertical tangents at: $(2, \pi)$, $\left(\dfrac{1}{2}, \dfrac{\pi}{3}\right)$, & $\left(\dfrac{1}{2}, \dfrac{5\pi}{3}\right)$

Singular points $\left(\dfrac{dy}{d\theta} = 0 \text{ \& } \dfrac{dx}{d\theta} = 0\right)$

There is a singular point at: $(0, 0)$

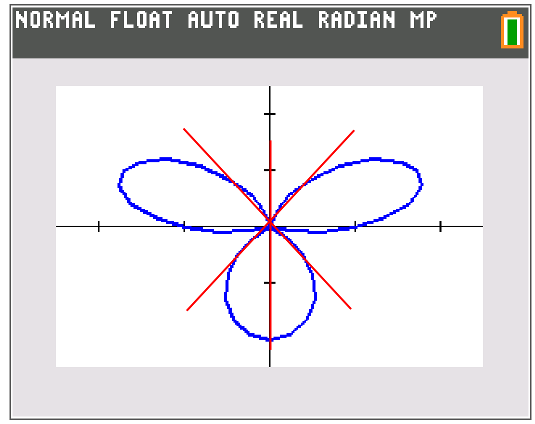
Example 4: The three-petal rose $r = \sin(3\theta) + 1$ has three tangent lines at the origin. Find the polar equation which represents each of these lines.

Any line tangent to the origin must go through the point $(0, \theta)$. Therefore $r = 0$ and we must solve the equation: $0 = \sin(3\theta) + 1$

$$0 = \sin(3\theta) + 1 \quad \Leftrightarrow \quad -1 = \sin(3\theta)$$

$$\Leftrightarrow \quad 3\theta = \frac{3\pi}{2} + 2\pi k \quad \Leftrightarrow \quad \theta = \frac{\pi}{2} + \frac{2\pi}{3}k$$

$$\text{So } \theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$



We note that in polar graphs $\theta = a$ is the equation of a line since the angle is fixed and the radius can take on any value.

Arc Length of Polar Curves

Motivation: We now consider the arc length of a polar curve.

Suppose we wish to find the length of a polar curve $r = f(\theta)$ where $a \le \theta \le b$. We can view the curve in terms of parametric equations as:

$$x = f(\theta)\cos(\theta) \quad y = f(\theta)\sin(\theta) .$$

We know from section 7.2 that the desired length will be given by: $L = \int_a^b \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta$. We can simplify this formula quite a bit by doing the following calculation:

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \left(\frac{dr}{d\theta}\cos(\theta) - r\sin(\theta)\right)^2 + \left(\frac{dr}{d\theta}\sin(\theta) + r\cos(\theta)\right)^2 = r^2 + \left(\frac{dr}{d\theta}\right)^2 .$$

This leads to the following formula.

Formula: The length of a curve with polar equation $r = f(\theta)$ where $a \le \theta \le b$, is

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta .$$

Example 5 (you try):

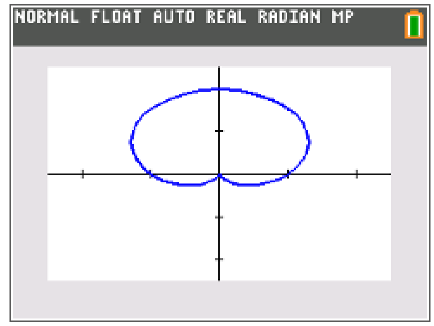
(a) Find the length of one tracing of the cardioid given by $r = 1+\sin(\theta)$. You may use a calculator to approximate the value of the definite integral.

First, we can see from the calculator and the function that the curve will be completely traced for $0 \le \theta \le 2\pi$.

We calculate that $\frac{dr}{d\theta} = \cos(\theta)$. Now, we apply the formula that we derived above and obtain:

$$L = \int_0^{2\pi} \sqrt{(1+\sin(\theta))^2 + \cos^2(\theta)}\ d\theta = \int_0^{2\pi} \sqrt{1+2\sin(\theta)+\sin^2(\theta)+\cos^2(\theta)}\ d\theta$$

$$= \int_0^{2\pi} \sqrt{2+2\sin(\theta)}\ d\theta = 8 .$$



(b) How would the problem change if you were working with function $r = \sin(\theta)$?

This function is traced exactly once for $0 \le \theta \le \pi$, so we would have the integral:

$$L = \int_0^{\pi} \sqrt{(\sin(\theta))^2 + \cos^2(\theta)}\, d\theta = \int_0^{\pi} \sqrt{1}\, d\theta = \pi$$

Example 6 (you try):

Find the *exact* length of the cardioid given by $r = e^{\theta}$ where $0 \le \theta \le \pi$. You may use a calculator to approximate the value of the definite integral.

First, we note that $\frac{dr}{d\theta} = e^{\theta}$. Now, we apply the formula that we derived above and obtain:

$$L = \int_0^{\pi} \sqrt{(e^{\theta})^2 + (e^{\theta})^2}\ d\theta = \int_0^{\pi} \sqrt{2e^{2\theta}}\ d\theta = \int_0^{\pi} \sqrt{2}e^{\theta}\ d\theta$$

$$= \sqrt{2}e^{\theta}\Big|_0^{\pi} = \sqrt{2}\left(e^{\pi}-1\right) \approx 31.3$$